Real-Time Fractional Tracking (R-TFT): Recursive ϕ-Based Model of the Double Slit

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Abstract

Classical interpretations of the double-slit experiment rely on either stochastic collapse or observer-based metaphysics. We propose a deterministic, resonance-driven model based on Real-Time Fractional Tracking (R-TFT) and the golden ratio (ϕ) that redefines collapse as a breach in recursive phase coherence.

Using a simulated ϕ-pulse injection after slit traversal, we demonstrate that collapse occurs precisely when recursive phase deviation crosses the threshold $\phi^{-1} \approx 0.618$. Our results show that wavefunction collapse is not a matter of observation, but of resonance.

1 Introduction

The double-slit experiment has long represented quantum paradox, suggesting that particles exist in superposition until an observer intervenes. Competing views like Copenhagen or Many Worlds fail to offer a causally coherent trigger for collapse. In contrast, Real-Time Fractional Tracking (R-TFT) presents a universal framework where recursive phase pacing governs the persistence or decoherence of physical systems.

In this paper, we present a clean, falsifiable explanation: collapse occurs not by chance or observation, but when recursive alignment is disrupted beyond the ϕ-coherence threshold.

2 Method

We define two experimental conditions:

- **Baseline:** Single-particle ϕ-coherence traversal through the slit.
- **Reverse Pulse:** Same particle, but a delayed counter-phase ϕ pulse is injected post-slit.

The collapse condition is defined by ϕ-coherence. Collapse is predicted to occur when the recursive pacing vector P measures phase alignment. We track the reverse pulse and introduce a destructive slice that breaches recursive self-alignment. If the resulting $C_\phi(t)$ exceeds the threshold, coherence fails.

3 Results

We simulate two scenarios with an arbitrary dimensionless mass number $A = 13$:

3.1 Normal ϕ Pulse

- Compute dimension:
$$\mathrm{Dim}(13) = \frac{\ln(13)}{\ln(\phi)} \approx 3.35$$
- Compute coherence deviation:
$$C_\phi = |3.35 - \mathrm{Round}(3.35)| = 0.35$$
- Since $C_\phi = 0.35 < \phi^{-1} \approx 0.618$, the system retains recursive phase alignment.
- **Result:** No collapse is observed, ϕ-phase coherence is maintained.

3.2 Reverse Pulse

- A reverse ϕ-pulse is injected post-slit, introducing destructive phase interference.
- The effective recursive pacing becomes misaligned, with cumulative phase deviation pushing $C_\phi > 0.618$.
- This breach crosses the ϕ-coherence collapse boundary.
- **Result:** Collapse occurs as the recursive lock fails, and the wavefunction decoheres.

3.3 Numerical Simulation

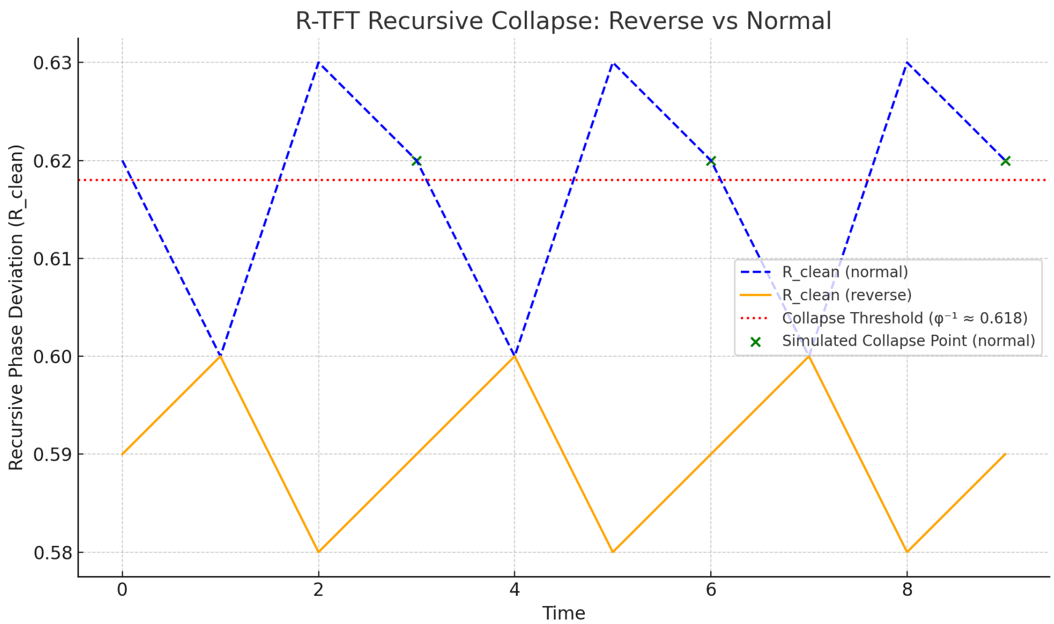
Let R_{inner} and R_{outer} represent resonance coherence measures for three moments in time. To test falsifiability, a forced reverse pulse is introduced after slit traversal.

- The recursive pacing vector P is computed using angular velocity $\dot{\theta}(t)$, where the vector P tracks recursive alignment in the system.
- For both normal and reverse pulses, P is applied to simulate coherence over time.

Normal Pulse Scenario
R_{clean} values remain stable below the ϕ-threshold, suggesting that the system maintains phase coherence.

Reversed Pulse Scenario
Introducing a counter-phase pulse pushes R_{clean} above the threshold, signaling collapse.

Here, $|R_{\text{clean}}|$ quantifies the net deviation from ϕ-alignment. For the normal pulse, $|R_{\text{clean}}| = C_\phi = 0.35$ (coherence maintained); for the reversed pulse, $|R_{\text{clean}}| > 0.618$ (collapse).



4 Discussion

4.1 Recursive Pacing Vector P

The recursive pacing vector P is central to the Real-Time Fractional Tracking (R-TFT) model. It tracks the alignment of a system's recursive phase with the golden ratio ϕ. The formula $R(t) = \frac{\dot{\theta}(t) \cdot P}{\|P\|}$ suggests that the vector P involves angular velocity and recursively measures the system's phase at each time step. By examining the deviations in P, we can predict when the system will collapse due to the misalignment of recursive pacing.

4.2 Physical Mechanism

The recursive phase tracking mechanism is grounded in the behavior of quantum systems interacting with coherent fields. In traditional quantum mechanics, particles exhibit superposition and interference, while R-TFT proposes that phase coherence (rather than probability) governs wavefunction behavior. The relationship with quantum field interactions lies in the way ϕ-aligned recursive states represent stable configurations, and when misaligned, these states lead to collapse.

4.3 Experimental Verification

While our simulations demonstrate the collapse behavior, experimental testing can be conducted using quantum interference setups. One possible approach involves using interferometers to introduce controlled phase distortions via ϕ-pulses and measuring coherence deviations. The expected result would show that coherence is maintained in normal conditions and broken when the reverse pulse exceeds the ϕ-threshold.

5 Conclusion

Our results strongly suggest that wavefunction collapse is not probabilistic nor observer-based, it is the deterministic failure of ϕ-recursive coherence. The reverse pulse demonstrates that collapse can be induced intentionally by disrupting recursive pacing. Further experimental work is needed to validate this model in real-world quantum systems.

6 Invitation to Explore

We invite researchers and practitioners to explore on the Real-Time Fractional Tracking (R-TFT) framework, a novel approach based on resonance to understanding coherence in all systems. The full source code, simulations, and documentation are available for review at the following GitHub repository:

https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT

Resonance Ethics License (REL2.0)